BT5240 - Computational Systems Biology

K. Varunkumar (BE23B016)

Question 1

a) The differential equations governing each of the substrates is given below:

$$R1: \quad \frac{-d[G]}{dt} = \frac{d[Py]}{dt} = k_1[G]$$

$$R2: \quad \frac{-d[Py]}{dt} = \frac{d[L]}{dt} = k_2[Py]$$

$$R3: \quad \frac{-d[G]}{dt} = \frac{d[W]}{dt} = k_3[G]^2$$

$$i) \quad \frac{d[G]}{dt} = -k_1[G] - k_3[G]^2$$

$$ii) \quad \frac{d[Py]}{dt} = k_1[G] - k_2[Py]$$

$$iii) \quad \frac{dL}{dt} = k_2[Py]$$

$$iv) \quad \frac{dW}{dt} = k_3[G]^2$$

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b) k1 =1.0 min^{-1}; k2 = 0.5 min^{-1}; k3 = 0.2 L.mol^{-1}.min^{-1}

Glucose = 1.0 mol/L; Pyruvate = Lactate = Waste products = 0 mol/L

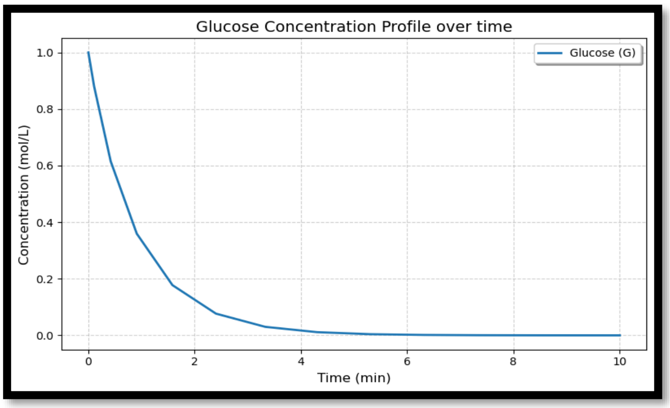

Figure 1.b.1: Glucose Concentration Profile

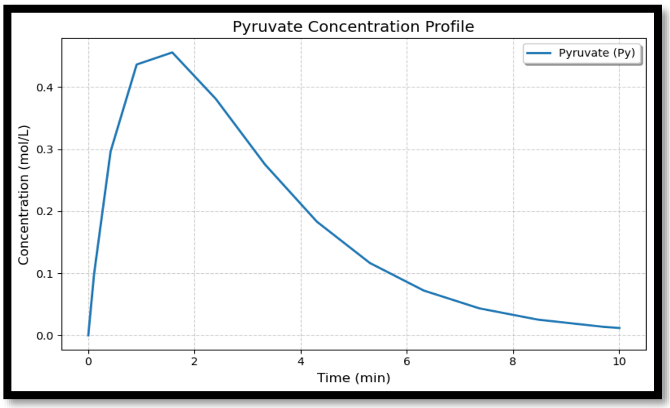

Figure 1.b.2: Pyruvate Concentration Profile

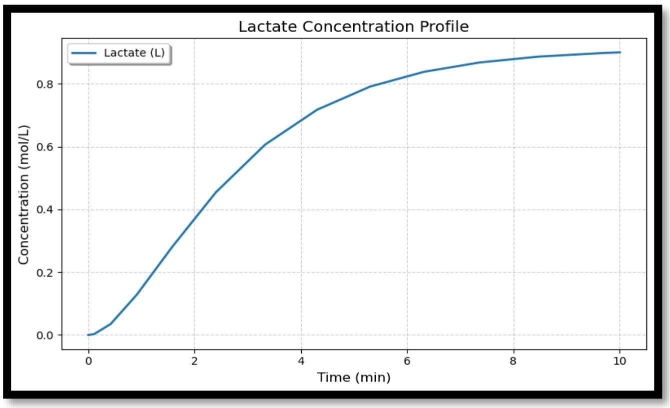

Figure 1.b.3: Lactate Concentration Profile

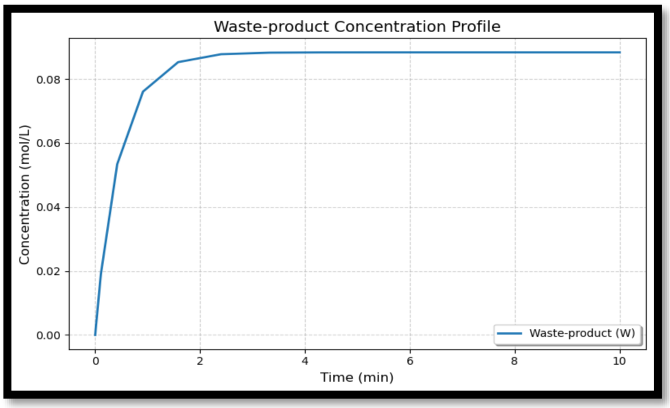

Figure 1.b.4: Waste-product Concentration Profile

c) The inferences drawn from the Figures in Question (b) are given below:

1. We find that [G] shows a hyperbolic decay ([G] is inversely proportional to time) as seen in Figure 1.b.1. This makes sense given that in this system glucose is only consumed and not replenished.
2. Pyruvate [Py] initial increases, with a sharp ascent phase (till time is approximately 1.8 minutes), when [G] is high and [Py] is low. However, once [G] falls substantially (time greater than 1.8 minutes) and [Py] nears its maximum value [Py] begins to gradually reduce (as seen in Figure 1.b.2).
3. The rate of lactate formation depends on [Py], being low at the start and end due to low [Py], and highest in the middle when [Py] is high (as seen in Figure 1.b.3).
4. The initial rate of waste-product formation is high due to high [G]. Around when [Py] begins to peak (which is to when most of the glucose is consumed, approximately t = 1.8 minutes), [W] begins to saturate (as seen in Figure 1.b.4). This is consistent the fact that waste-product acts as a sink in our system, leading to saturation as glucose is progressively consumed.

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d) Assuming R1 follows Michaelis-Menten kinetics, differential equations governing each of the substrates is given below (K_m =0.5 mol/L; v_{max} =1.2 mol $L^{-1}min^{-1}$):

$$R1: \quad \frac{-d[G]}{dt} = \frac{d[Py]}{dt} = \frac{V_{max} * [G]}{K_m + [G]}$$

$$R2: \quad \frac{-d[Py]}{dt} = \frac{d[L]}{dt} = k_2[Py]$$

$$R3: \quad \frac{-d[G]}{dt} = \frac{d[W]}{dt} = k_3[G]^2$$

$$i) \quad \frac{d[G]}{dt} = -\frac{V_{max} * [G]}{K_m + [G]} - k_3[G]^2$$

$$ii) \quad \frac{d[Py]}{dt} = \frac{V_{max} * [G]}{K_m + [G]} - k_2[Py]$$

$$iii) \quad \frac{dL}{dt} = k_2[Py]$$

$$iv) \quad \frac{dW}{dt} = k_3[G]^2$$

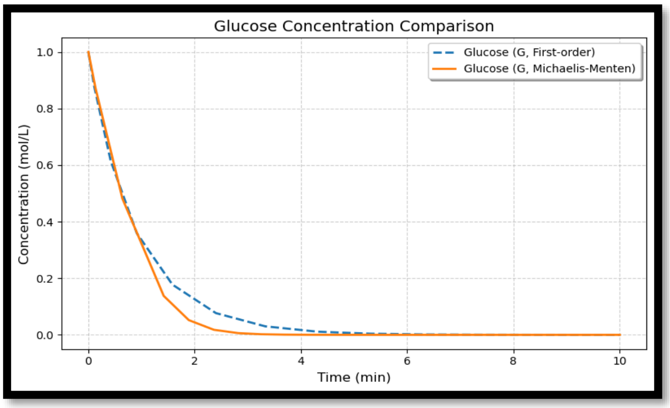

Figure 1.d.1: Glucose Concentration Profile

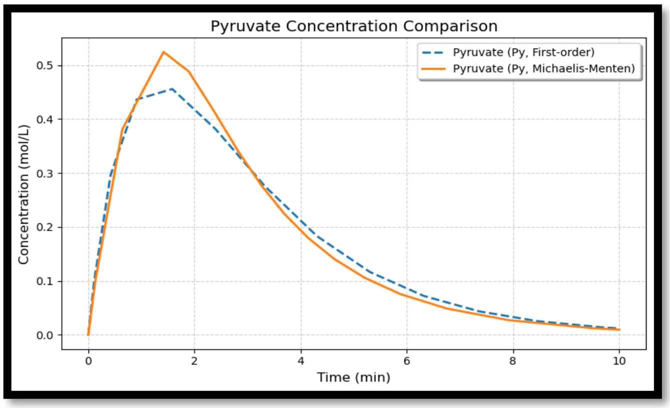

Figure 1.d.2: Pyruvate Concentration Profile

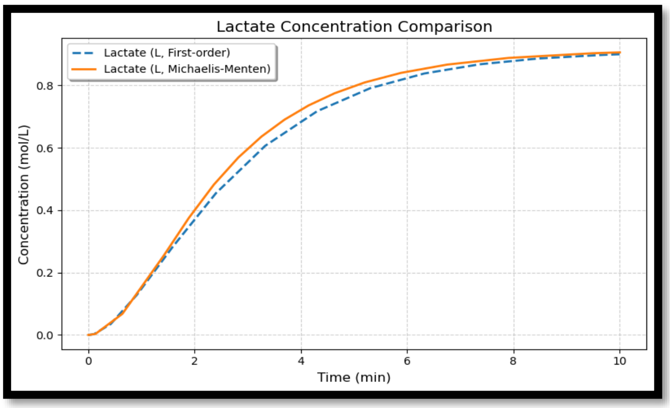

Figure 1.d.3: Lactate Concentration Profile

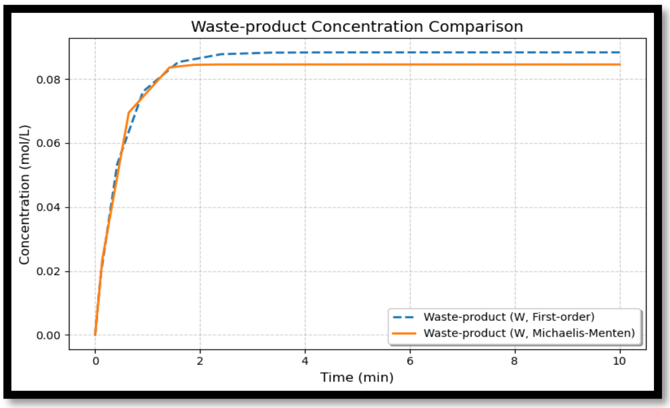

Figure 1.d.4: Waste-product Concentration Profile

From Figures 1.d.1 through 1.d.4, we see that there is a difference in the initial rate of the reactions between both the models. However, over time, both models eventually converge to a fixed steady state.

The initial rate in the Michaelis-Menten model is greater during the for the initial duration. Although this is not always true in general (as we see in part e), it occurs in this case due to the chosen K_m value.

The reason why both model's rates and steady state concentrations appear to converge is due to the fact that $V = V_{max} * [G]/ (K_m + [G])$, and at low [G] (or $K_m >> [G]$) this reaction practically becomes a first-order rate equation. Thus, after t = 2 minutes, both models show similar kinetics/trajectories.

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e) The system was simulated for different values of K_m (0.5 mol/L, 0.2 mol/L, 1.0 mol/L, and 2.0 mol/L).

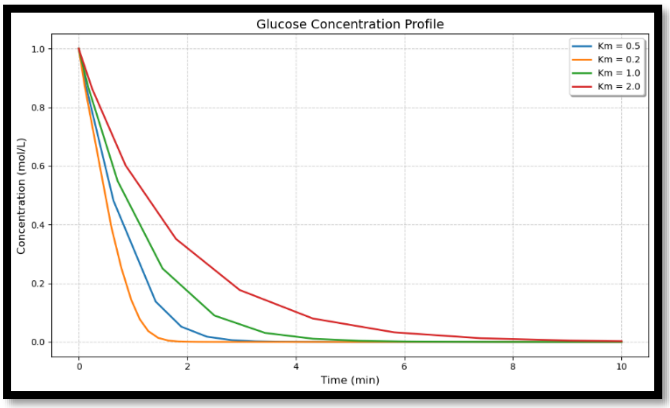

Figure 1.e.1: Glucose Concentration Profile

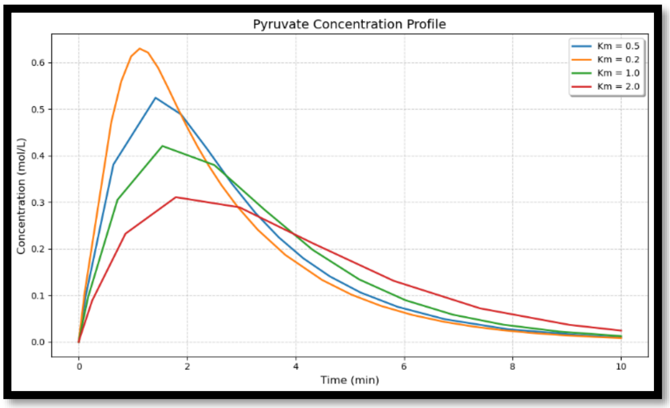

Figure 1.e.2: Pyruvate Concentration Profile

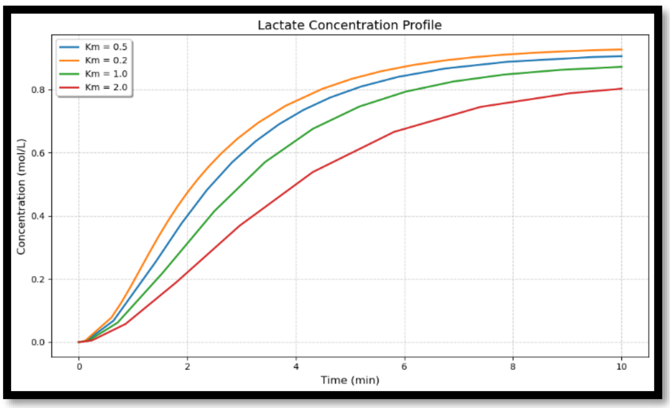

Figure 1.e.3: Lactate Concentration Profile

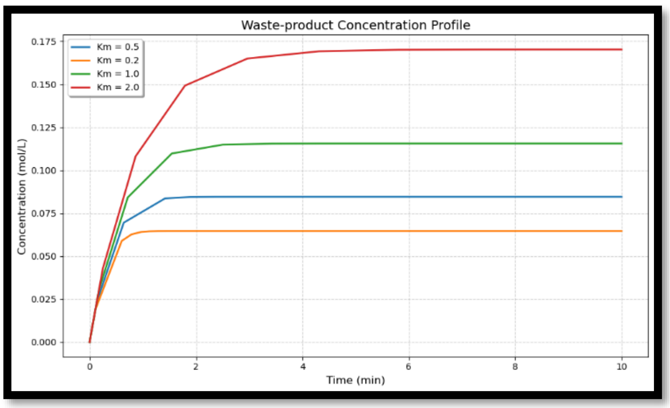

Figure 1.e.4: Waste-product Concentration Profile

As K_m decreases, the rate of breakdown of glucose increases, as seen in Figure 1.e.1.

Consequently, pyruvate and waste-product concentrations reach their peaks earlier (as seen in Figures 1.e.2, 1.e.4 respectively), while the final lactate concentration increases (as seen in Figure 1.e.3).

We see that the ratio of Lactate : Waste-product increases, showing that Glucose => Pyruvate => Lactate pathway (primarily Glucose => Pyruvate) is favoured over Glucose => Waste-product. This intuitively illustrates the fact that K_m controls the affinity of an enzyme to a substrate (when K_m increases, we see the reverse trend play out, with Glucose => Waste-product being the more preferred pathway).

The trends discussed above are reversed when K_m increases.

Question 2

Assumption: For this problem we assume synchronous update of the Boolean Network.

a) The given regulatory network is represented using the Bipartite graph shown below:

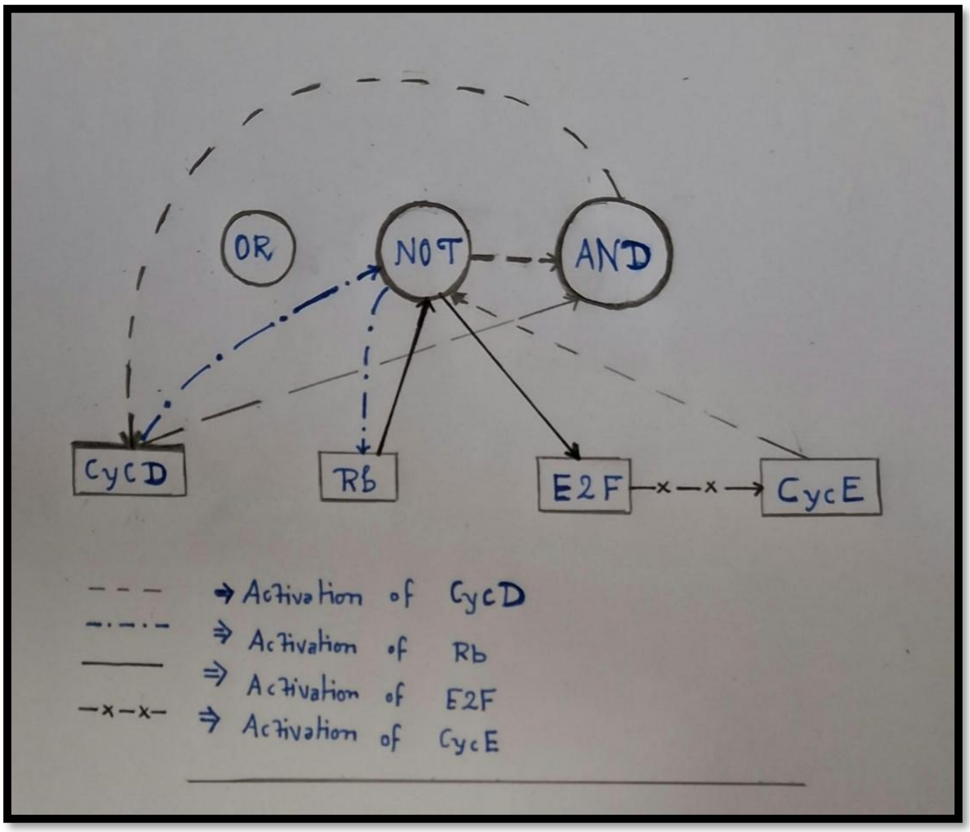

Figure 2.a.1

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b) The rules to update the Boolean network substrates are given below:

I) CycD (t+1) = CycD (t) and (not CycE (t))

II) Rb (t+1) = not CycD (t)

III) E2F (t+1) = not Rb (t)

IV) CycE (t+1) = E2F (t)

c) The state transition matrix for all possible initial states is shown below:

S.No	CycD (t)	Rb (t)	E2F (t)	CycE (t)	CycD (t+1)	Rb (t+1)	E2F (t+1)	CycE (t+1)
1	0	0	0	0	0	1	1	0
2	0	0	0	1	0	1	1	0
3	0	0	1	0	0	1	1	1
4	0	0	1	1	0	1	1	1
5	0	1	0	0	0	1	0	0
6	0	1	0	1	0	1	0	0
7	0	1	1	0	0	1	0	1
8	0	1	1	1	0	1	0	1
9	1	0	0	0	1	0	1	0
10	1	0	0	1	0	0	1	0
11	1	0	1	0	1	0	1	1
12	1	0	1	1	0	0	1	1
13	1	1	0	0	1	0	0	0
14	1	1	0	1	0	0	0	0
15	1	1	1	0	1	0	0	1
16	1	1	1	1	0	0	0	1

The state transition diagram represented by the above table is shown below:

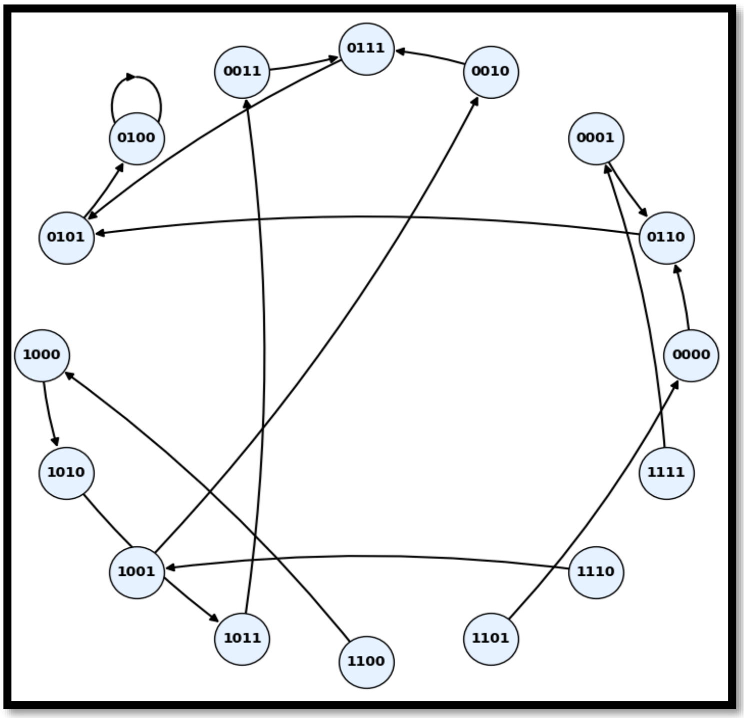

Figure 2.c.1: State Transition Diagram

d) The system was simulated from all possible initial states for 10 iterations and the results are shown below:

i) Initial State: (0, 0, 0, 0) t=0: (0, 0, 0, 0) t=1: (0, 1, 1, 0) t=2: (0, 1, 0, 1) t=3: (0, 1, 0, 0) t=4: (0, 1, 0, 0) t=5: (0, 1, 0, 0) t=6: (0, 1, 0, 0) t=7: (0, 1, 0, 0) t=8: (0, 1, 0, 0) t=9: (0, 1, 0, 0) t=10: (0, 1, 0, 0)	ii) Initial State: (0, 0, 0, 1) t=0: (0, 0, 0, 1) t=1: (0, 1, 1, 0) t=2: (0, 1, 0, 1) t=3: (0, 1, 0, 0) t=4: (0, 1, 0, 0) t=5: (0, 1, 0, 0) t=6: (0, 1, 0, 0) t=7: (0, 1, 0, 0) t=8: (0, 1, 0, 0) t=9: (0, 1, 0, 0) t=10: (0, 1, 0, 0)
iii) Initial State: (0, 0, 1, 0) t=0: (0, 0, 1, 0) t=1: (0, 1, 1, 1) t=2: (0, 1, 0, 1) t=3: (0, 1, 0, 0) t=4: (0, 1, 0, 0) t=5: (0, 1, 0, 0) t=6: (0, 1, 0, 0) t=7: (0, 1, 0, 0) t=8: (0, 1, 0, 0) t=9: (0, 1, 0, 0) t=10: (0, 1, 0, 0)	iv) Initial State: (0, 0, 1, 1) t=0: (0, 0, 1, 1) t=1: (0, 1, 1, 1) t=2: (0, 1, 0, 1) t=3: (0, 1, 0, 0) t=4: (0, 1, 0, 0) t=5: (0, 1, 0, 0) t=6: (0, 1, 0, 0) t=7: (0, 1, 0, 0) t=8: (0, 1, 0, 0) t=9: (0, 1, 0, 0) t=10: (0, 1, 0, 0)
v) Initial State: (0, 1, 0, 0) t=0: (0, 1, 0, 0) t=1: (0, 1, 0, 0) t=2: (0, 1, 0, 0) t=3: (0, 1, 0, 0) t=4: (0, 1, 0, 0) t=5: (0, 1, 0, 0) t=6: (0, 1, 0, 0) t=7: (0, 1, 0, 0) t=8: (0, 1, 0, 0) t=9: (0, 1, 0, 0)	vi) Initial State: (0, 1, 0, 1) t=0: (0, 1, 0, 1) t=1: (0, 1, 0, 0) t=2: (0, 1, 0, 0) t=3: (0, 1, 0, 0) t=4: (0, 1, 0, 0) t=5: (0, 1, 0, 0) t=6: (0, 1, 0, 0) t=7: (0, 1, 0, 0) t=8: (0, 1, 0, 0) t=9: (0, 1, 0, 0)

t=10: (0, 1, 0, 0)	t=10: (0, 1, 0, 0)
vii) Initial State: (0, 1, 1, 0) t=0: (0, 1, 1, 0) t=1: (0, 1, 0, 1) t=2: (0, 1, 0, 0) t=3: (0, 1, 0, 0) t=4: (0, 1, 0, 0) t=5: (0, 1, 0, 0) t=6: (0, 1, 0, 0) t=7: (0, 1, 0, 0) t=8: (0, 1, 0, 0) t=9: (0, 1, 0, 0) t=10: (0, 1, 0, 0)	viii) Initial State: (0, 1, 1, 1) t=0: (0, 1, 1, 1) t=1: (0, 1, 0, 1) t=2: (0, 1, 0, 0) t=3: (0, 1, 0, 0) t=4: (0, 1, 0, 0) t=5: (0, 1, 0, 0) t=6: (0, 1, 0, 0) t=7: (0, 1, 0, 0) t=8: (0, 1, 0, 0) t=9: (0, 1, 0, 0) t=10: (0, 1, 0, 0)
ix) Initial State: (1, 0, 0, 0) t=0: (1, 0, 0, 0) t=1: (1, 0, 1, 0) t=2: (1, 0, 1, 1) t=3: (0, 0, 1, 1) t=4: (0, 1, 1, 1) t=5: (0, 1, 0, 1) t=6: (0, 1, 0, 0) t=7: (0, 1, 0, 0) t=8: (0, 1, 0, 0) t=9: (0, 1, 0, 0) t=10: (0, 1, 0, 0)	x) Initial State: (1, 0, 0, 1) t=0: (1, 0, 0, 1) t=1: (0, 0, 1, 0) t=2: (0, 1, 1, 1) t=3: (0, 1, 0, 1) t=4: (0, 1, 0, 0) t=5: (0, 1, 0, 0) t=6: (0, 1, 0, 0) t=7: (0, 1, 0, 0) t=8: (0, 1, 0, 0) t=9: (0, 1, 0, 0) t=10: (0, 1, 0, 0)
xi) Initial State: (1, 0, 1, 0) t=0: (1, 0, 1, 0) t=1: (1, 0, 1, 1) t=2: (0, 0, 1, 1) t=3: (0, 1, 1, 1) t=4: (0, 1, 0, 1) t=5: (0, 1, 0, 0) t=6: (0, 1, 0, 0) t=7: (0, 1, 0, 0) t=8: (0, 1, 0, 0) t=9: (0, 1, 0, 0) t=10: (0, 1, 0, 0)	xii) Initial State: (1, 0, 1, 1) t=0: (1, 0, 1, 1) t=1: (0, 0, 1, 1) t=2: (0, 1, 1, 1) t=3: (0, 1, 0, 1) t=4: (0, 1, 0, 0) t=5: (0, 1, 0, 0) t=6: (0, 1, 0, 0) t=7: (0, 1, 0, 0) t=8: (0, 1, 0, 0) t=9: (0, 1, 0, 0) t=10: (0, 1, 0, 0)

xiii) Initial State: (1, 1, 0, 0) t=0: (1, 1, 0, 0) t=1: (1, 0, 0, 0) t=2: (1, 0, 1, 0) t=3: (1, 0, 1, 1) t=4: (0, 0, 1, 1) t=5: (0, 1, 1, 1) t=6: (0, 1, 0, 1) t=7: (0, 1, 0, 0) t=8: (0, 1, 0, 0) t=9: (0, 1, 0, 0) t=10: (0, 1, 0, 0)	xiv) Initial State: (1, 1, 0, 1) t=0: (1, 1, 0, 1) t=1: (0, 0, 0, 0) t=2: (0, 1, 1, 0) t=3: (0, 1, 0, 1) t=4: (0, 1, 0, 0) t=5: (0, 1, 0, 0) t=6: (0, 1, 0, 0) t=7: (0, 1, 0, 0) t=8: (0, 1, 0, 0) t=9: (0, 1, 0, 0) t=10: (0, 1, 0, 0)
xv) Initial State: (1, 1, 1, 0) t=0: (1, 1, 1, 0) t=1: (1, 0, 0, 1) t=2: (0, 0, 1, 0) t=3: (0, 1, 1, 1) t=4: (0, 1, 0, 1) t=5: (0, 1, 0, 0) t=6: (0, 1, 0, 0) t=7: (0, 1, 0, 0) t=8: (0, 1, 0, 0) t=9: (0, 1, 0, 0) t=10: (0, 1, 0, 0)	xvi) Initial State: (1, 1, 1, 1) t=0: (1, 1, 1, 1) t=1: (0, 0, 0, 1) t=2: (0, 1, 1, 0) t=3: (0, 1, 0, 1) t=4: (0, 1, 0, 0) t=5: (0, 1, 0, 0) t=6: (0, 1, 0, 0) t=7: (0, 1, 0, 0) t=8: (0, 1, 0, 0) t=9: (0, 1, 0, 0) t=10: (0, 1, 0, 0)

This system appears to have a single attractor at (0,1,0,0), with all initial states converging to this final state.

e) The evolution of system variables in all the above cases is plotted and shown below:

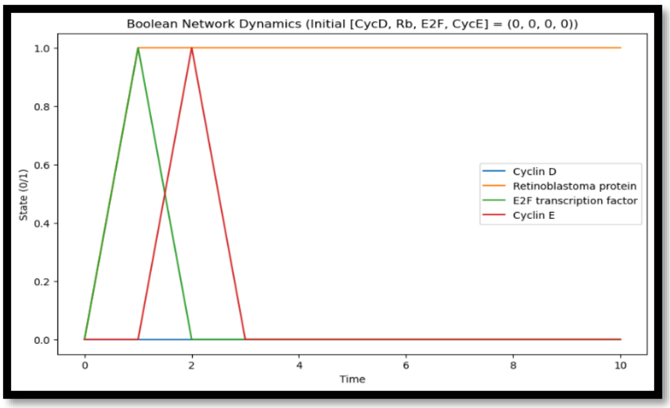

Figure 2.e.1: Initial State: (0, 0, 0, 0)

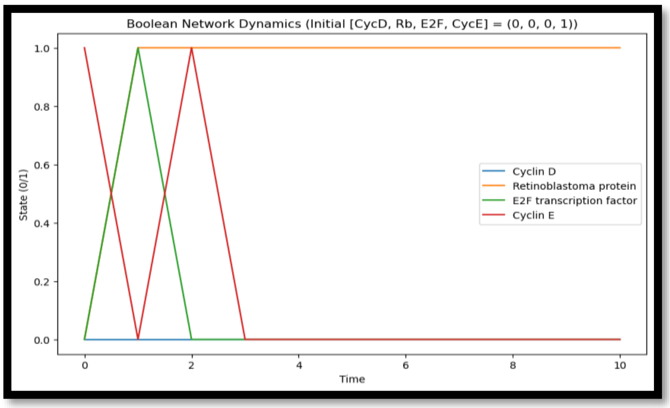

Figure 2.e.2: Initial State: (0, 0, 0, 1)

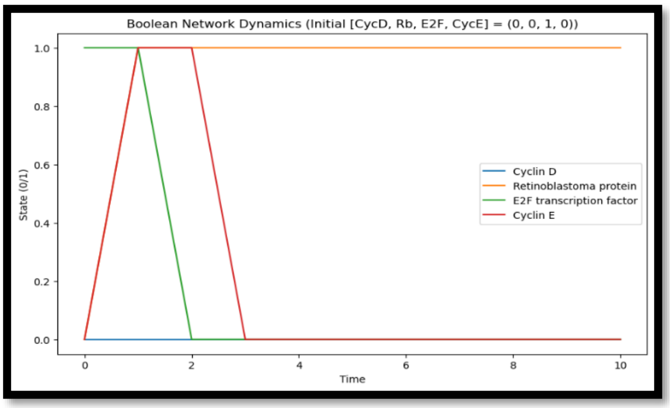

Figure 2.e.3: Initial State: (0, 0, 1, 0)

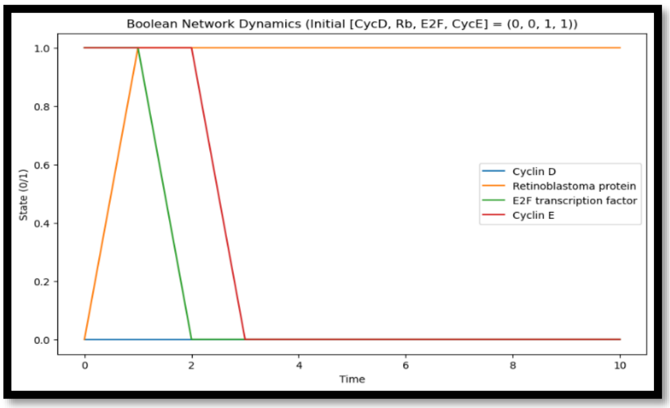

Figure 2.e.4: Initial State: (0, 0, 1, 1)

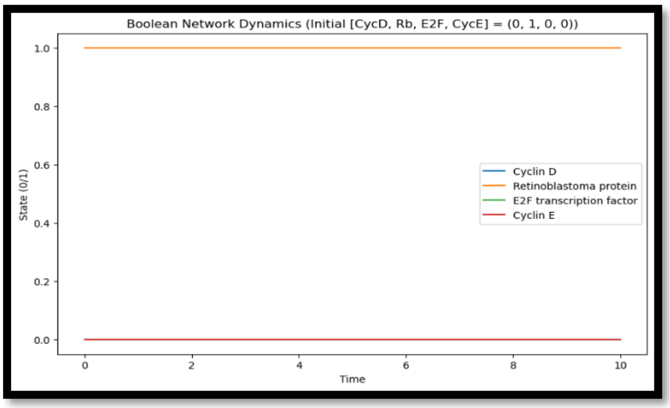

Figure 2.e.5: Initial State: (0, 1, 0, 0)

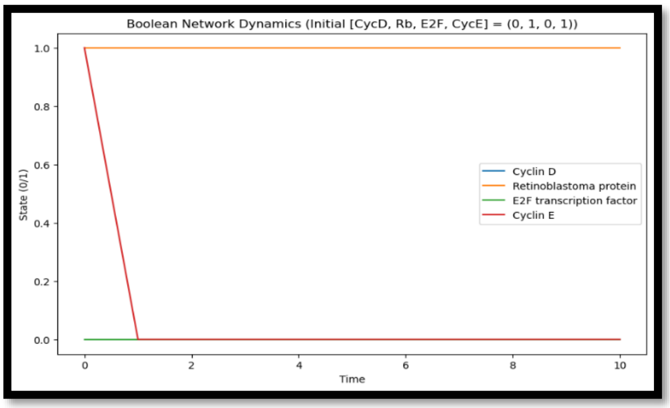

Figure 2.e.6: Initial State: (0, 1, 0, 1)

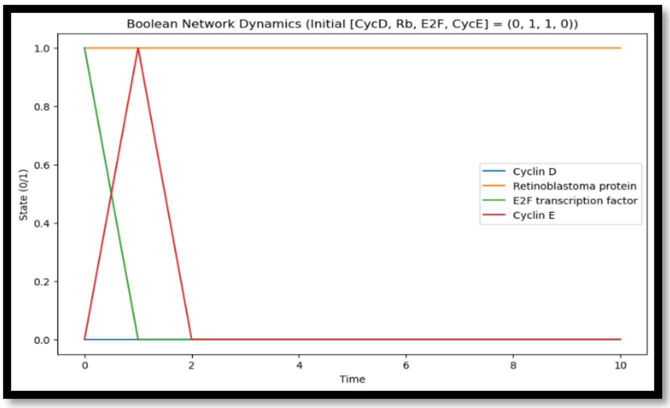

Figure 2.e.7: Initial State: (0, 1, 1, 0)

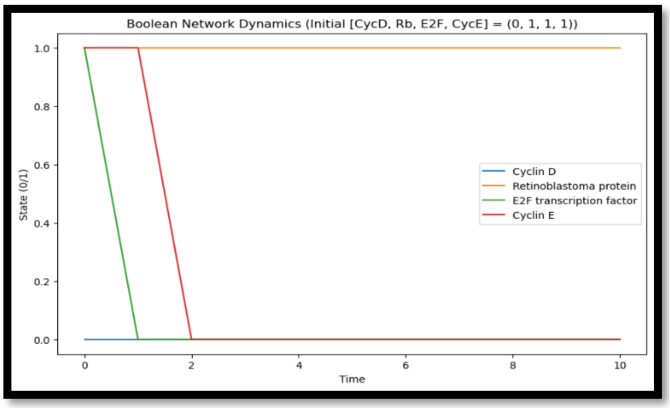

Figure 2.e.8: Initial State: (0, 1, 1, 1)

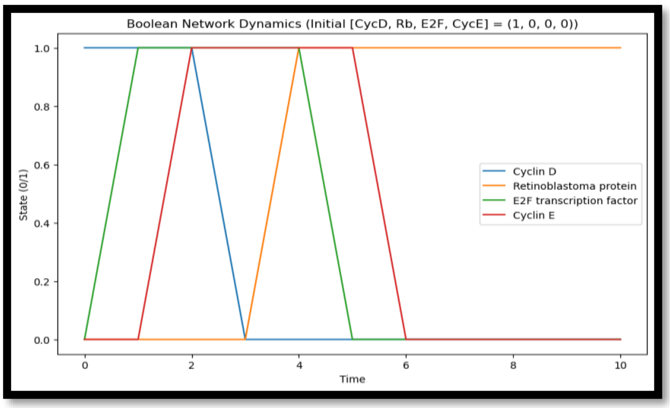

Figure 2.e.9: Initial State: (1, 0, 0, 0)

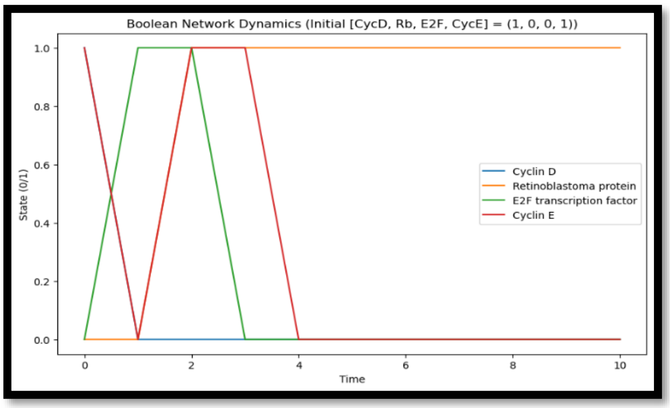

Figure 2.e.10: Initial State: (1, 0, 0, 1)

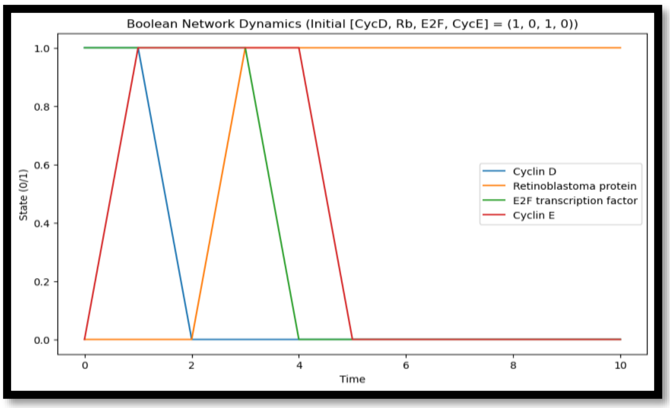

Figure 2.e.11: Initial State: (1, 0, 1, 0)

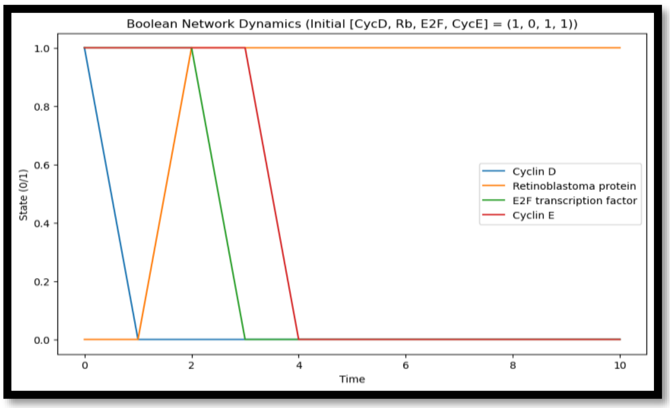

Figure 2.e.12: Initial State: (1, 0, 1, 1)

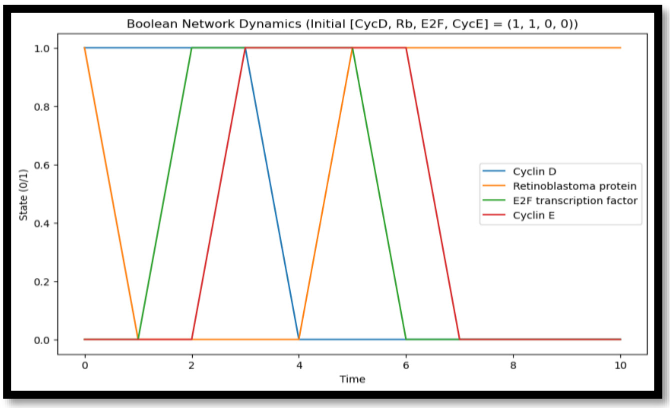

Figure 2.e.13: Initial State: (1, 1, 0, 0)

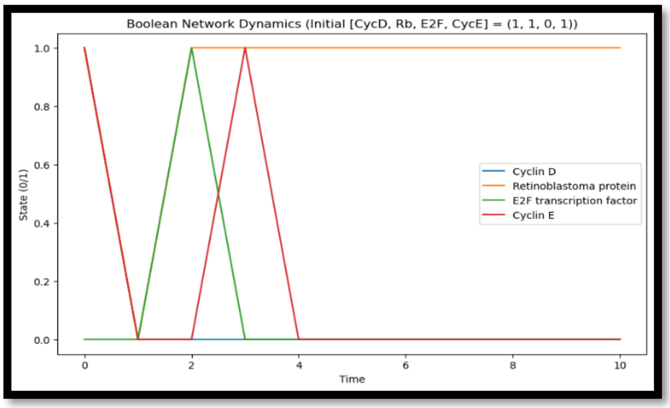

Figure 2.e.14: Initial State: (1, 1, 0, 1)

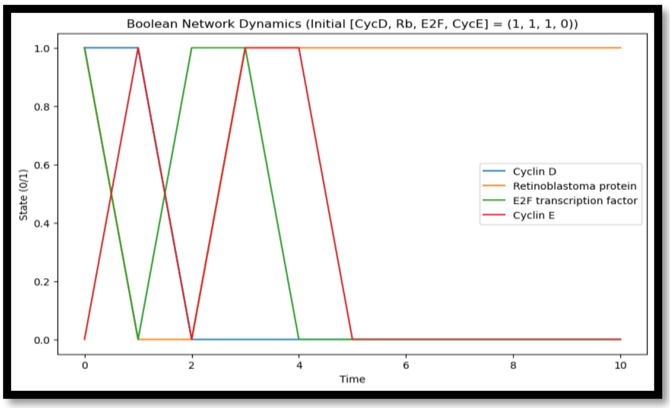

Figure 2.e.15: Initial State: (1, 1, 1, 0)

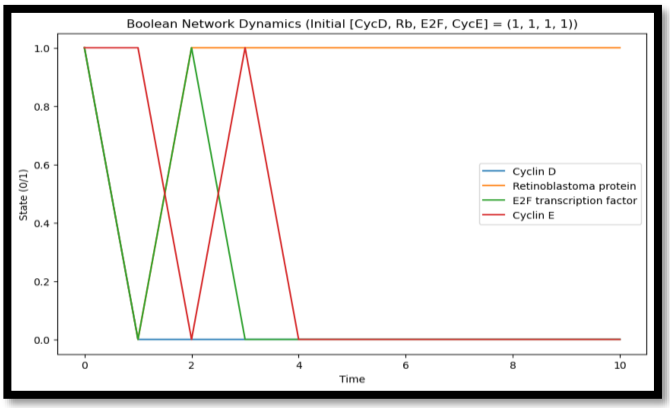

Figure 2.e.16: Initial State: (1, 1, 1, 1)

In Figures 2.e.1 to 2.e.16, the state of each component is represented by: Blue (Cyclin D), Orange (Retinoblastoma protein), Green (E2F transcription factor), Red (Cyclin E).

We see that the attractor (stable state of the system) is indeed at (0, 1, 0, 0), with rapid convergence to this state in most cases.

From a biological perspective this Boolean network represents a system which rapidly returns to the steady state in the presence of a transient perturbation. The longest time taken by the system to reach the steady state is 7 iterations from the state (1, 1, 0, 0). Thus, we qualitatively make the argument that this system models short-term cellular response to stress.